

Microsoft Entra ID Multifactor Authentication

Project type: Hands-on IAM security lab **Date:** October 8, 2026

Environment: Microsoft Entra ID tenant, test identity (Alex Chen), Microsoft Authenticator, Microsoft Edge InPrivate, Entra sign-in logs

Project overview

Configured and validated authentication methods for a test identity in Microsoft Entra ID. Investigated an interrupted registration flow, tested an Authenticator number-matching challenge, and reviewed sign-in activity to validate that an MFA requirement was satisfied.

Objectives

1. Confirm registered MFA methods for a test user.
2. Initiate and test a sign-in requiring multifactor authentication.
3. Investigate authentication interruptions and failures.
4. Validate the sign-in result through Microsoft Entra sign-in logs.

Implementation and troubleshooting

- 1. Authentication methods:** The test identity had Microsoft Authenticator and a software OATH token listed as usable methods. The default sign-in method was an OATH TOTP code; system-preferred MFA was enabled with phone app notification.
- 2. Registration interruption:** A security-method setup flow displayed a timeout/general error. Instead of assuming MFA registration failed, the administrator checked the user's registered methods.
- 3. Challenge test:** A sign-in prompted an Authenticator approval with number matching. This establishes that an MFA challenge was presented; the numeric value is intentionally not treated as a reusable secret.
- 4. Log investigation:** The interactive sign-in log showed unsuccessful and interrupted events before a successful event. The successful event was inspected to confirm its MFA requirement and result.

Evidence 1 — Authenticator number-matching challenge

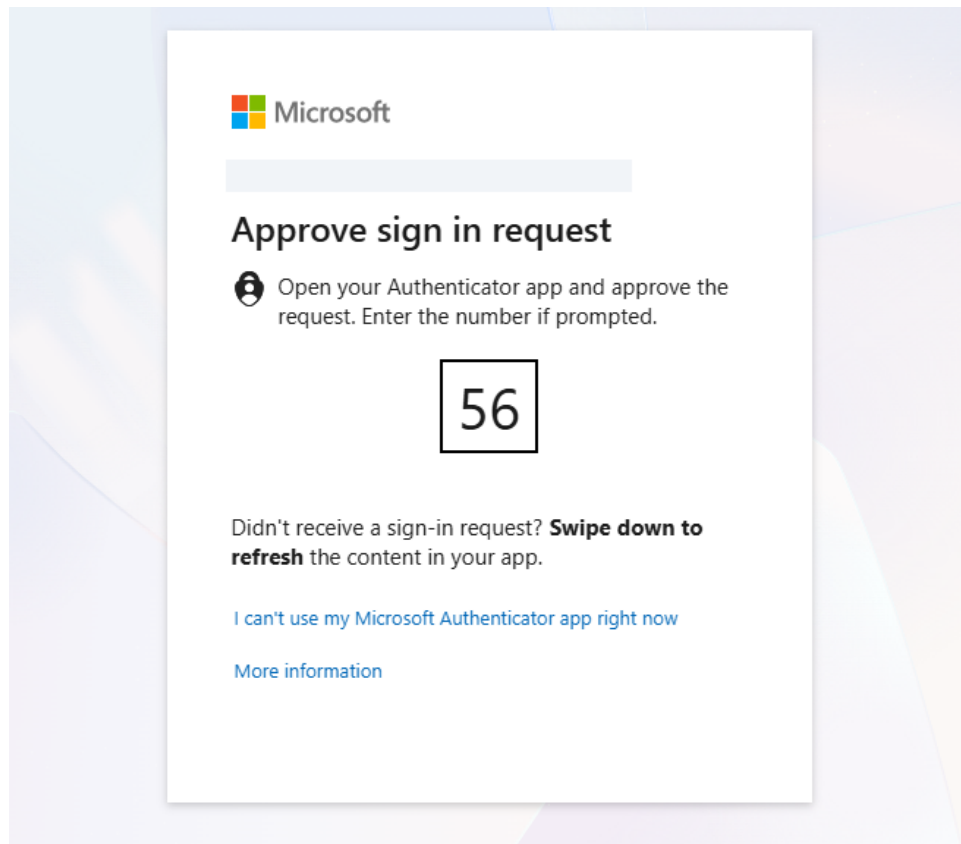


Figure 1. Microsoft sign-in prompted the test identity to approve a number-matching request. Account identifier redacted.

Sign-in evidence

Evidence 2 — Interactive sign-in events

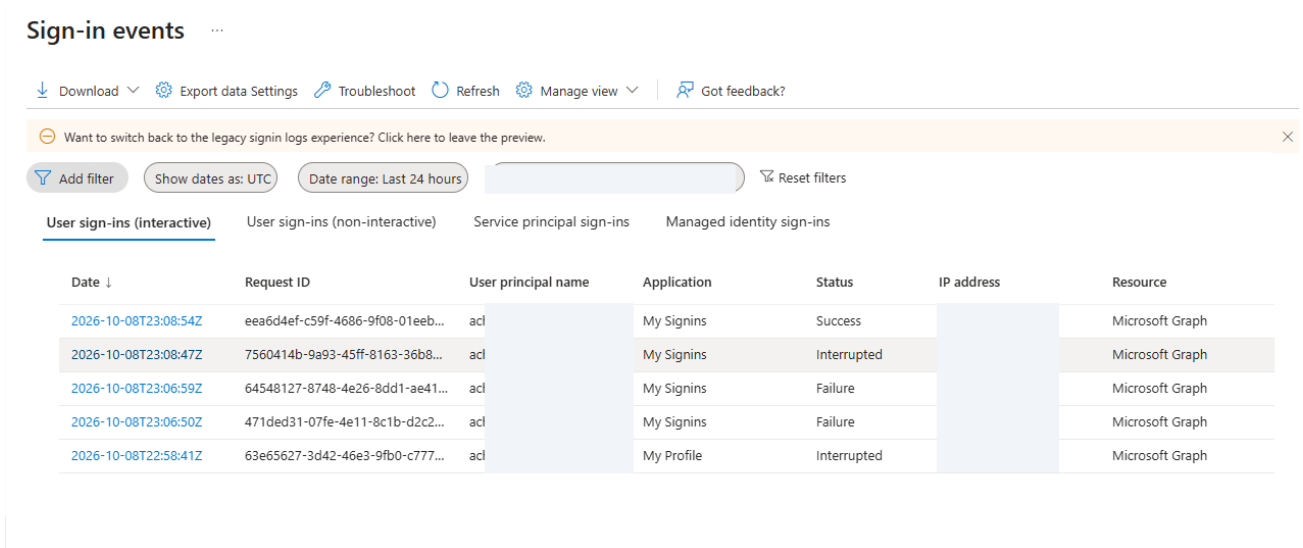


Figure 2. Interactive sign-in events show earlier failure/interruption records followed by success. User principal names and IP addresses redacted.

Evidence 3 — Successful MFA-required sign-in

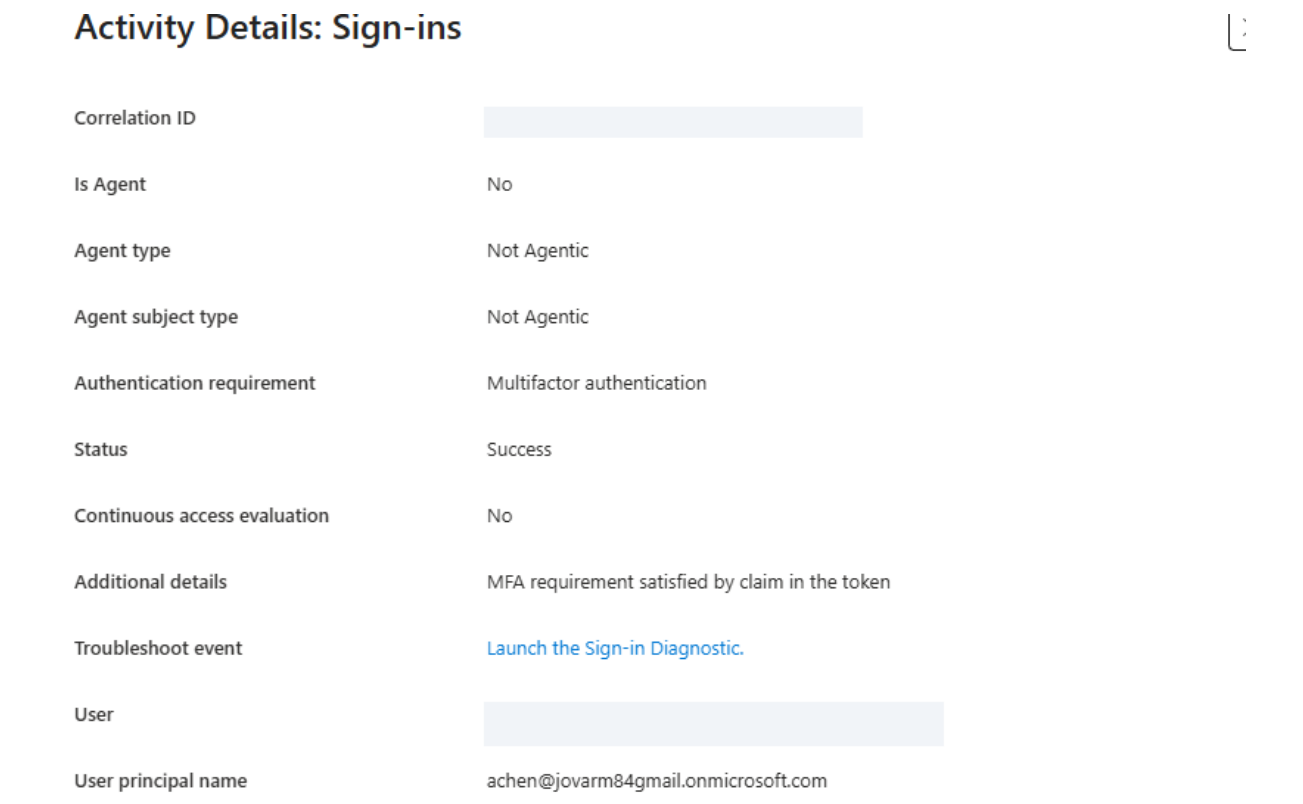


Figure 3. Activity details show “Authentication requirement: Multifactor authentication,” “Status: Success,” and “MFA requirement satisfied by claim in the token.” Sensitive identifiers redacted.

Authentication validation & outcome

Evidence 4 — Authentication details

Activity Details: Sign-ins

Basic info	Location	Device info	Authentication details	Conditional access	Report only	...
Authentication policies applied		Security Defaults				
Date	Authentication...	Authentication...	Succeeded	Result detail	Requirement	
2026-10-08T23:08:54Z	Previously satis...		Yes	MFA requireme...		

Figure 4. The authentication details show Security Defaults applied and a successful result with a previously satisfied MFA claim.

Results and interpretation

Check	Observed outcome
Authenticator registration	Registered and usable
Number-matching prompt	Presented during sign-in
Interactive sign-in	Success recorded
MFA requirement	Satisfied by existing claim in token
Policy shown	Security Defaults

Security interpretation

The successful event confirms that the MFA requirement was met. Because the log says **“previously satisfied”** / **“satisfied by claim in the token,”** it does **not** establish that a fresh Authenticator prompt was completed in that same event. The separate number-matching screenshot documents that a challenge was presented.

Skills demonstrated

Microsoft Entra ID • MFA registration review • Microsoft Authenticator • Number matching • Security Defaults • Sign-in log analysis • Identity troubleshooting • Security evidence documentation

Portfolio note

This project uses a fictional test identity in a controlled lab. Screenshots are sanitized for public portfolio sharing. No production organization or employer system is represented.